413/613 Fall 2026 - Applied Statistics & Data Science I - Lecture 11

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RECALL

In the previous lecture, we considered different ways of estimating an unknown parameter θ from observed data

$$\mathbf{x} = (x_1, \ldots, x_n).$$

Maximum likelihood estimation uses only the observed data:

$$\hat{\theta}_{\text{MLE}} = \arg\max_{\theta} p(\mathbf{x} \mid \theta).$$

If we also have prior information about plausible parameter values, we can introduce a prior distribution $p(\theta)$. Bayes' rule gives

$$p(\theta \mid \mathbf{x}) = \frac{p(\mathbf{x} \mid \theta)p(\theta)}{p(\mathbf{x})}.$$

The maximum a posteriori estimate is

$$\hat{\theta}_{\text{MAP}} = \arg\max_{\theta} p(\theta \mid \mathbf{x}) = \arg\max_{\theta} p(\mathbf{x} \mid \theta)p(\theta).$$

Thus,

$$\text{MLE: data} \longrightarrow \hat{\theta}_{\text{MLE}},$$

$$\text{MAP: data + prior} \longrightarrow \hat{\theta}_{\text{MAP}}.$$

Bayesian inference goes one step further. Rather than reducing our information about θ to a single point estimate, we retain the entire posterior distribution

$$\boxed{p(\theta \mid \mathbf{x}).}$$

The natural next question is:

$$\boxed{\text{How does this distribution change as new observations arrive?}}$$

1 Bayesian updating

The posterior distribution is

$$p(\theta \mid \mathbf{x}) = \frac{p(\mathbf{x} \mid \theta)p(\theta)}{p(\mathbf{x})}.$$

The denominator

$$p(\mathbf{x}) = \int p(\mathbf{x} \mid \theta)p(\theta)\, d\theta$$

does not depend on θ; its role is to normalize the posterior so that it integrates to one. Therefore, when studying the dependence on θ, we often write

$$\boxed{p(\theta \mid \mathbf{x}) \propto p(\mathbf{x} \mid \theta)p(\theta).}$$

Thus, Bayesian inference has a simple interpretation:

$$\boxed{\text{posterior} \propto \text{likelihood} \times \text{prior}.}$$

The likelihood describes what the observed data tell us about θ, while the prior describes what we believed about θ before observing those data. Their combination produces the posterior distribution.

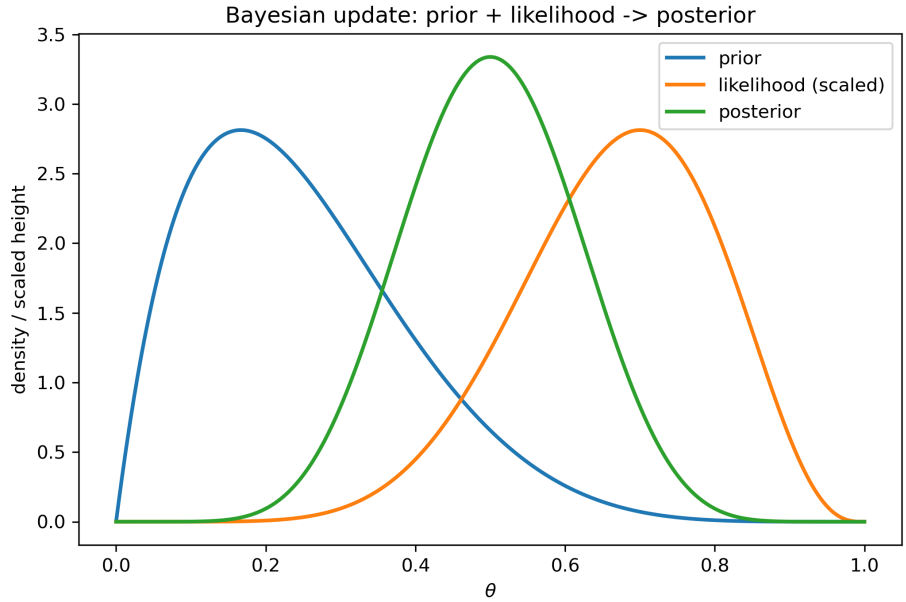


Figure 1: Schematic illustration of a Bayesian update. The prior distribution $p(\theta)$ represents our knowledge about the parameter before observing the data, while the likelihood $p(\mathbf{x} \mid \theta)$ describes which parameter values are most compatible with the observed data. Their product, after normalization, gives the posterior distribution $p(\theta \mid \mathbf{x}) \propto p(\mathbf{x} \mid \theta)p(\theta)$.

1.1 Sequential Bayesian updating

Suppose our initial uncertainty about θ is described by

$$p_0(\theta).$$

After observing a new data point x_1,

$$p_1(\theta) = p(\theta \mid x_1) \propto p(x_1 \mid \theta)p_0(\theta).$$

The posterior $p_1(\theta)$ now contains everything we have learned from the first observation. When a second observation x_2 arrives, this posterior can be used as the prior for the next update:

$$p_2(\theta) \propto p(x_2 \mid \theta)p_1(\theta).$$

More generally, after observing

$$x_1, \ldots, x_k,$$

define

$$p_k(\theta) = p(\theta \mid x_1, \ldots, x_k).$$

When a new observation x_{k+1} arrives,

$$\boxed{p_{k+1}(\theta) \propto p(x_{k+1} \mid \theta)p_k(\theta).}$$

Thus the Bayesian update can be viewed as the repeated operation

$$\boxed{\text{prior} \longrightarrow \text{multiply by likelihood of new data} \longrightarrow \text{posterior} \longrightarrow \text{new prior.}}$$

If the observations are conditionally independent given θ, then repeated application of this rule gives

$$p(\theta \mid x_1, \ldots, x_n) \propto p(\theta) \prod_{i=1}^{n} p(x_i \mid \theta).$$

Therefore, processing the observations sequentially gives the same posterior as processing the entire dataset at once.

2 What are conjugate priors and why are they useful ?

Although the update rule is simple,

$$p(\theta \mid \mathbf{x}) \propto p(\mathbf{x} \mid \theta)p(\theta),$$

the resulting form of the posterior distribution may be difficult to calculate. A particularly convenient situation occurs when multiplying the likelihood by the prior produces a posterior belonging to the same family of distributions as the prior. Such a prior is called a **conjugate prior**.

$$\boxed{\text{prior family} \xrightarrow{\times \text{ likelihood}} \text{same posterior family.}}$$

This is useful because the posterior can immediately be used as the prior for the next observation without changing the form of the distribution. For example, suppose a coin has an unknown probability of heads

$$\theta \in [0, 1].$$

For n independent flips, if h heads are observed, the likelihood is

$$p(h \mid \theta) = \binom{n}{h} \theta^h (1-\theta)^{n-h}.$$

Consider a Beta prior,

$$\theta \sim \text{Beta}(\alpha, \beta),$$

with density

$$p(\theta) \propto \theta^{\alpha-1}(1-\theta)^{\beta-1}.$$

Multiplying prior and likelihood,

$$p(\theta \mid h) \propto \theta^h (1-\theta)^{n-h} \theta^{\alpha-1}(1-\theta)^{\beta-1}.$$

Collecting powers gives

$$p(\theta \mid h) \propto \theta^{\alpha+h-1}(1-\theta)^{\beta+n-h-1}.$$

Therefore,

$$\boxed{\theta \mid \mathbf{x} \sim \text{Beta}\left(\alpha + h,\ \beta + n - h\right).}$$

The posterior is again a Beta distribution. Hence the Beta distribution is conjugate to the Bernoulli/Binomial likelihood. If $t = n - h$ denotes the number of tails, the update is particularly simple:

$$\boxed{\alpha_{\text{new}} = \alpha_{\text{old}} + h, \qquad \beta_{\text{new}} = \beta_{\text{old}} + t.}$$

For a single new coin flip,

	α	β
Head	$+1$	0
Tail	0	$+1$

Thus, in this example, Bayesian updating reduces to keeping track of the number of observed heads and tails. As more observations are collected, the posterior typically becomes more concentrated, reflecting increasing information about the unknown parameter θ. Figure 2 illustrates this sequential update starting from a uniform prior,

$$\theta \sim \text{Beta}(1, 1).$$

After each new coin flip, the posterior remains a Beta distribution, with its parameters updated according to the accumulated numbers of heads and tails. Initially, with little data, the posterior is broad and our uncertainty about θ is large. As more observations are collected, the posterior becomes increasingly concentrated.

In the example shown, the data are generated from a fair coin with

$$\theta_{\text{true}} = 0.5.$$

Although the posterior may shift noticeably after only a few observations, the effect of individual coin flips becomes smaller as the sample size grows. With increasing data, the posterior concentrates around the true value $\theta = 0.5$, while its spread decreases.

Thus, the figure illustrates two important features of Bayesian updating:

$$\boxed{\text{more data} \quad \Longrightarrow \quad \text{less sensitivity to individual observations}}$$

and

$$\boxed{\text{more data} \quad \Longrightarrow \quad \text{smaller posterior uncertainty.}}$$

What happens as more data are observed? Initially, the posterior may depend strongly on the prior. As more observations are collected, the likelihood typically becomes increasingly informative. Schematically,

$$p(\theta) \quad \longrightarrow \quad p(\theta \mid x_1) \quad \longrightarrow \quad p(\theta \mid x_1, x_2) \quad \longrightarrow \quad \cdots$$

and the posterior often becomes increasingly concentrated around parameter values that are consistent with the observed data. Thus, Bayesian inference does not only provide a point estimate. It gives a distribution that changes as evidence accumulates, allowing us to quantify how our uncertainty about θ changes with the amount of data.

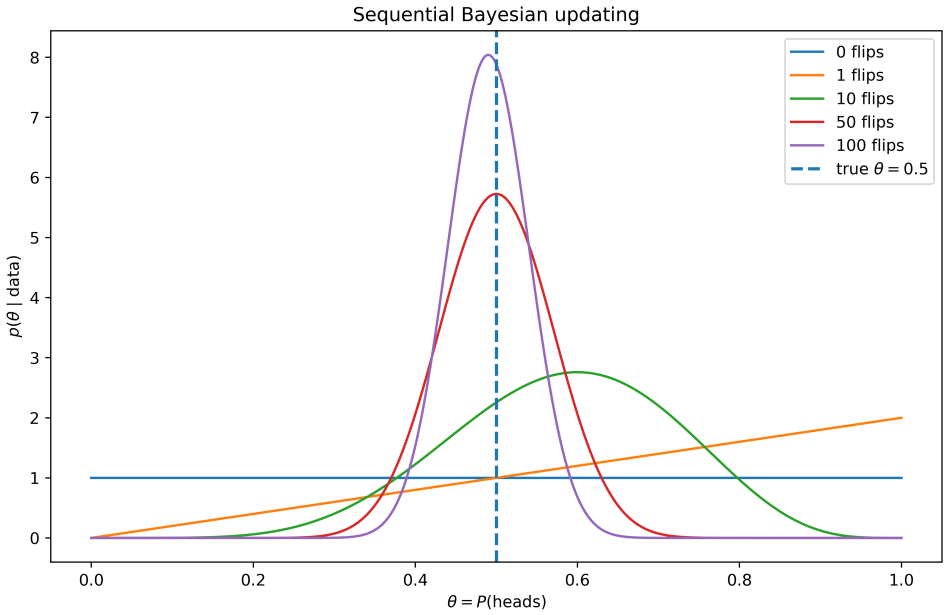


Figure 2: Sequential Bayesian updating for the unknown probability of heads θ. Starting from the uniform prior Beta$(1, 1)$, each observed head increments α and each observed tail increments β. The posterior remains in the Beta family and becomes increasingly concentrated as more coin flips are observed. In this example, the data are generated from a fair coin with $\theta_{\text{true}} = 0.5$.

The Beta distribution is a very flexible distribution for parameters constrained to the interval $0 \leq \theta \leq 1$. This makes it particularly useful for probabilities such as the probability of heads of a coin.

We write

$$\theta \sim \text{Beta}(\alpha, \beta), \qquad \alpha > 0, \quad \beta > 0,$$

with density

$$\boxed{p(\theta) = \frac{1}{B(\alpha, \beta)} \theta^{\alpha-1}(1-\theta)^{\beta-1}, \qquad 0 \leq \theta \leq 1.}$$

Here, $B(\alpha, \beta)$ is a normalization constant chosen so that

$$\int_0^1 p(\theta)\, d\theta = 1.$$

The parameters α and β control the shape of the distribution. Its mean is

$$\boxed{\mathbb{E}[\theta] = \frac{\alpha}{\alpha + \beta}.}$$

For example, $\text{Beta}(1, 1)$ is the uniform distribution on $[0, 1]$, representing no preference for any particular value of θ. If $\alpha = \beta$ and both are larger than one, the distribution is symmetric and concentrated around $\theta = \frac{1}{2}$. Thus, the Beta family provides a flexible way of representing different prior beliefs about an unknown probability θ.

3 Bayesian inference as hypothesis testing

So far, Bayesian inference has been used to estimate and quantify the uncertainty about an unknown parameter θ through the posterior distribution

$$p(\theta \mid \mathbf{x}).$$

Suppose now that we are not considering a continuum of possible values of θ, but only a small number of competing possibilities. For example, suppose a coin is known to be either

$$H_1 : \theta = 0.5, \qquad H_2 : \theta = 0.7,$$

where θ is the probability of observing heads. The unknown quantity is therefore no longer an arbitrary value of θ, but which of the two hypotheses is correct:

$$H \in \{H_1, H_2\}.$$

Bayesian inference can be applied in exactly the same way. Before observing any data, we assign prior probabilities

$$P(H_1), \qquad P(H_2),$$

representing our initial uncertainty about the two competing models. After observing data $\mathbf{x}$, Bayes' rule gives

$$\boxed{P(H_j \mid \mathbf{x}) = \frac{P(\mathbf{x} \mid H_j) P(H_j)}{\sum_\ell P(\mathbf{x} \mid H_\ell) P(H_\ell)}.}$$

Here,

$$P(\mathbf{x} \mid H_j)$$

is the likelihood of the observed data under hypothesis H_j, while

$$P(H_j \mid \mathbf{x})$$

is the posterior probability that hypothesis H_j is the correct model. Thus, Bayesian hypothesis testing asks:

After observing the data, how should our probability assigned to each competing hypothesis change?

Example For the coin example, suppose we begin with equal prior probabilities,

$$P(H_1) = P(H_2) = 0.5,$$

where

$$H_1 : \theta_1 = 0.5, \qquad H_2 : \theta_2 = 0.7.$$

For a single new observation $x_k \in \{0, 1\}$, where

$$x_k = \begin{cases} 1, & \text{heads}, \\ 0, & \text{tails}, \end{cases}$$

the likelihood under hypothesis H_j is

$$\boxed{P(x_k \mid H_j) = \theta_j^{x_k}(1 - \theta_j)^{1-x_k}.}$$

Thus, under H_1,

$$P(x_k \mid H_1) = 0.5^{x_k}(0.5)^{1-x_k},$$

while under H_2,

$$P(x_k \mid H_2) = 0.7^{x_k}(0.3)^{1-x_k}.$$

For example, if the first observation is a head, $x_1 = 1$, then

$$P(x_1 = 1 \mid H_1) = 0.5, \qquad P(x_1 = 1 \mid H_2) = 0.7.$$

Using Bayes' rule,

$$P(H_j \mid x_1) = \frac{P(x_1 \mid H_j)P(H_j)}{P(x_1 \mid H_1)P(H_1) + P(x_1 \mid H_2)P(H_2)}.$$

Therefore,

$$P(H_1 \mid x_1 = 1) = \frac{0.5 \times 0.5}{0.5 \times 0.5 + 0.7 \times 0.5} = \frac{5}{12} \approx 0.417,$$

and

$$P(H_2 \mid x_1 = 1) = \frac{0.7 \times 0.5}{0.5 \times 0.5 + 0.7 \times 0.5} = \frac{7}{12} \approx 0.583.$$

Thus, observing a head shifts probability toward H_2, since a head is more likely under $\theta = 0.7$ than under $\theta = 0.5$. If instead the first observation is a tail, $x_1 = 0$, then

$$P(x_1 = 0 \mid H_1) = 0.5, \qquad P(x_1 = 0 \mid H_2) = 0.3,$$

and therefore the posterior shifts toward H_1. The key idea is that each observation is weighted according to how probable that observation is under each competing hypothesis.

We know from Bayesian update that the posterior probabilities become the priors for the next observation:

$$P_1(H_j) = P(H_j \mid x_1).$$

After observing x_2,

$$P(H_j \mid x_1, x_2) = \frac{P(x_2 \mid H_j)P(H_j \mid x_1)}{\sum_\ell P(x_2 \mid H_\ell)P(H_\ell \mid x_1)}.$$

More generally, after observing $x_1, \ldots, x_k$,

$$\boxed{P_k(H_j) = P(H_j \mid x_1, \ldots, x_k),}$$

and after observing the next data point x_{k+1},

$$\boxed{P_{k+1}(H_j) = \frac{P(x_{k+1} \mid H_j)P_k(H_j)}{\sum_\ell P(x_{k+1} \mid H_\ell)P_k(H_\ell)}.}$$

Thus, as new coin flips are observed,

$$P(H_1), P(H_2) \longrightarrow P(H_1 \mid x_1), P(H_2 \mid x_1) \longrightarrow P(H_1 \mid x_1, x_2), P(H_2 \mid x_1, x_2) \longrightarrow \cdots$$

Figure 3 shows this sequential updating process. Because the data in the example are generated from a fair coin, the posterior probabilities fluctuate substantially at first, when only a few observations have been collected. Individual heads can temporarily increase support for H_2, while tails tend to increase support for H_1. As more evidence accumulates, however, these fluctuations become less important and the posterior probability increasingly concentrates on

$$H_1 : \theta = 0.5.$$

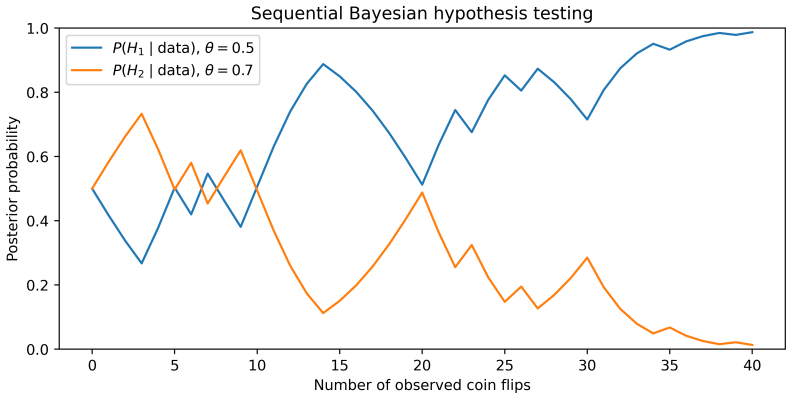


Figure 3: Sequential Bayesian hypothesis testing for two candidate coin models, $H_1 : \theta = 0.5$ and $H_2 : \theta = 0.7$, starting from equal prior probabilities. Each coin flip contributes a likelihood under both hypotheses, and Bayes' rule combines these likelihoods with the current prior probabilities to produce updated posterior probabilities. These posteriors then become the priors for the next observation. Since the data are generated from a fair coin, accumulated evidence eventually favors H_1.

4 Advantages and limitations of Bayesian inference

Bayesian inference provides a principled way of combining observed data with prior information and produces a full probability distribution over the unknown parameter. This can be particularly useful when the dataset is small or when meaningful prior information is available. However, these advantages also introduce several difficulties.

Sensitivity to the prior. The posterior distribution $p(\theta|\mathbf{x})$ depends on the choice of the prior $p(\theta)$. When the amount of data is small, different reasonable priors can lead to substantially different posterior distributions. Therefore, the prior should be regarded as part of the statistical model and its influence should be examined carefully.

Choice of prior. In some problems, meaningful prior information may be available from previous experiments, physical constraints, or scientific knowledge. In other problems, the choice of prior may be less clear and can introduce subjective modeling decisions.

Computational cost. The posterior is

$$p(\theta \mid \mathbf{x}) = \frac{p(\mathbf{x} \mid \theta)p(\theta)}{p(\mathbf{x})},$$

where

$$p(\mathbf{x}) = \int p(\mathbf{x} \mid \theta)p(\theta)\, d\theta.$$

For simple models, this integral and the resulting posterior may be available analytically. For more complicated or high-dimensional models, however, the posterior may be difficult to compute directly. In such cases, numerical methods such as Monte Carlo sampling are often needed. Thus, Bayesian inference replaces the problem of finding a single parameter estimate with the richer, but potentially more difficult, problem of characterizing an entire probability distribution over the parameters.